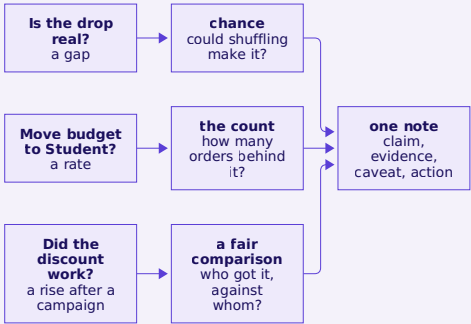


1 Three questions, three checks, one note



A gap gets a chance reference, a rate gets its count, and a campaign gets a fair comparison. Every check ends as one line of the same note.

Crux. A p-value is a share of chance-only worlds; it is never the chance the finding is wrong.

2 Choosing the route, chapter by chapter

Question	Best fit here	Second route	Switch when
Real or wobble?	Shuffle test	Textbook two-sample test	Thousands of well-behaved rows
Worth acting on?	Rupees against company and cost	Bootstrap range	A known recovery rate
Trust a rate?	Coin flips on its count	Count every deal	A cheap way to buy more orders
Did a discount work?	Split inside each segment	Both groups on one mix	Groups chosen by a coin
What goes to the CEO?	Four-part note, under 200 words	Trace every figure back	The same review every week
Did the sale cause it?	Random hold-back, next time	Chapter 4's split	A hold-back is impossible

3 The shuffle test, and the sentence it supports

<pre>random.seed(2026) for _ in range(5000): random.shuffle(pool) gaps.append(mean(pool[:k]) - mean(pool[k:])) share = sum(1 for g in gaps if g >= real_gap) / len(gaps)</pre>	
Draft	Survives Kavya
"A 3 percent chance we are wrong."	"If nothing had changed, a gap this large turns up in about 3 of every 100 shuffles."
"p = 0.34 proves nothing changed."	"The gap sits inside the usual wobble."

4 Real, then worth it

A small share says a gap beats chance, never that it is big: with enough orders even Rs 20 comes back significant. Size it per member, for the segment and against the company's quarter, then against the cost of acting: a fix that costs C against a fall of F pays only if it wins back more than C over F. A bootstrap range that dips below the cost means test on part of the group first.

Crux. Real and worth acting on are two separate calls: the shuffle answers the first, rupees against cost answer the second.

5 The count behind a rate

Orders behind the rate	One order moves it by
10	10 points
30	3.3 points
100	1 point
400	0.25 points

Flip a coin per order, thousands of times, and see how often chance alone makes the rise.

Crux. Count what a rate stands on before you repeat it; under thirty, it is a lead.

6 Who got it, against whom

A blend can rise while every segment falls, when the group that got the campaign holds more of the segment that spends more anyway. Compare inside each segment, state the mix of the two groups, and put both groups on one mix for a one-line answer. A discount of d needs volume to rise by 1 / (1 minus d), less one, just to stand still: 17.6 percent at 15 percent off.

Crux. Split an aggregate by segment before you credit a campaign, and name who got it.

7 The note to Meera

Claim: one sentence she can act on. Evidence: the number with its base and its count. Caveat: what would change the claim. Action: what to do next and what it costs. Audit each line: a base, a count or chance, a caveat.

Crux. The note is claim, evidence, caveat, action, and "not yet, and here is what would tell us" is a complete answer.

8 The fair comparison

Before and after credits the campaign with the month. Put an untargeted segment beside it, count the orders under each month, and decide the next campaign's groups with a coin, inside each segment, before it starts.

Crux. A fair comparison asks who got it, who did not, and what else changed; only a coin makes the two groups alike.

THE WORDS ON THIS SHEET

Label shuffle Dealing labels at random to build a chance-only world	p-value Share of chance-only worlds with a gap at least as large	Break-even recovery Share of a fall a fix must win back	Confidence interval The range of sizes the data supports	Confounder Drives both who got a campaign and the outcome
			Lead A rate on under thirty observations	